

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester Master of Technology (CE)

University Examination May 2011

CE703: Algorithm Complexity and Computations

Date: 11/05/2011

Time: 10:00 a.m to 1:00 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION-I

Q-1

- [A] Arrange following functions in n in increasing order:
 $2^n, \log_2 n, n^3, n^2 \log_2 n, 3^n$ 2
- [B] What will be the best case and worst case for Quick-sort algorithm? 2
- [C] How to choose an algorithm for solving a single problem, when you have multiple algorithms for solving same problem are available? 2
- [D] Insertion sort is better than selection sort in the case when array is given into reverse sorted order. Justify the Statement. 2
- [E] Write an algorithm to find the maximum element in an array A storing n integers. What is the running time of this algorithm for computing the maximum element in an array of integers? 3

Q-2

- [A] Consider the recurrence
 $T(n) = 4T(n/2) + n^2$ When n is power of 2, $n \geq 2$.
Solve the above recurrence without using master theorem. 6
- [B] What are the general characteristics of Greedy algorithms? 3
- [C] Illustrate the operation of HEAPSORT on the array $A = \{5, 13, 2, 25, 7, 17, 20, 8, 4\}$ 3

OR

Q-2

- [A] Let $S = \{a, b, c, d, e, f, g\}$ be a collection of objects with benefit-weight values as follows : a : (12,4), b: (10,6), c : (8,5), d : (11,7) , e: (14,3), f: (7,1), g: (9,6). What is an optimal solution to the fractional knapsack problem for S assuming the sack can hold objects with total weight 18? 4
- [B] What are the difference between greedy algorithm and dynamic programming? 4
- [C] Explain with the help of an example, 0/1 knapsack Problem using Dynamic Programming. 4

Q-3

- [A] Develop an algorithm using recursive and iterative function to calculate the Fibonacci Series of two integers A and B . Also Find the Best Case and Worst Case time complexity of both functions. 4
- [B] What is Asymptotic Notation? Explain O , Ω , θ notation with suitable example. 5
- [C] What do you mean by deterministic and Non Deterministic Algorithms? Differentiate between them. Write example for each of them. 3

OR

- [C] Solve using master Method:
 $T(n) = 2T(n/2) + n$
 $T(n) = 2T(n/2) + n^2$ 3

SECTION-II

Q-4

- [A] Differentiate between following graph representation. Adjacency Matrix and Adjacency List. 2
- [B] Explain General template of Backtracking Method. 2
- [C] What do you mean by polynomial time complexity and logarithmic complexity? Which one is higher? 2
- [D] Explain Greedy Choice Property with the help of an example. 2
- [E] Explain common characteristics of Dynamic Programming Problems. 3

Q-5

- [A] What is Backtracking? Generate at least 3-solutions for 5-Queen's problem. Draw a pruned state space tree for the problem. 5
- [B] Differentiate decision problem and optimization problem. 2
- [C] Assigning 4 people to 4 jobs so that the total cost is minimized. Each person does one job and each job is assigned to one person. Cost matrix is given below: 5

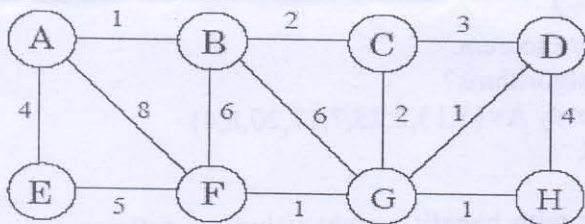
	Job 1	Job 2	Job 3	Job 4	
$C =$	9	2	7	8	Person a
	6	4	3	7	Person b
	5	8	1	8	Person c
	7	6	9	4	Person d

Use branch and bound to solve the assignment problem (Minimization) with the following cost matrices.

OR

Q-5

- [A] 5



[Fig : 1]

Find the shortest path using Dijkstra algorithm using Source as A.

- [B] Compare backtracking and branch-and-bound. 3
- [C] Define Binary Search. Explain How to solve binary search problem using divide and conquer technique. 4

Q-6

- [A] Write Floyd's Algorithm and explain with the help of an example. 4
- [B] What is preprocessing and matching time for naïve string matching algorithm? 4
- [C] Provide a divide-and-conquer algorithm for determining the smallest and second smallest values in a given unordered set of numbers. Provide a recurrence equation expressing the time complexity of the algorithm. 4

OR

- [C] Explain topological sort with example. 4